

An interpretable reduced-order model for real-time estimation of near-wall turbulence

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Abstract

The near-wall region of a turbulent flow maintains a self-sustaining cycle of streaks and quasi-streamwise vortices, and this cycle leaves an uneven footprint on the wall shear stress. Whether its dynamics, and not merely a snapshot, can be recovered from the wall shear alone is the question addressed here. A reduced-order model is fitted directly in the wall-shear coordinates: the trajectory is resampled uniformly in arc length to drive the empirical measure towards the physical one, the states are clustered under a local Mahalanobis metric adapted to the attractor's geometry, and a single global regression over radial basis functions gives a differentiable latent vector field. Applied to a minimal turbulent channel at friction Reynolds number 180, the geometry-adapted clustering separates the mature-streak, breakdown and quiescent phases into sharp populations that map onto the known coherent-structure skeleton, without physical labels. Integrated freely, the model stays bounded and reproduces the invariant measure; being differentiable, it locates the cycle's weakly unstable near-equilibria and resolves its non-normal growth and phase-volume contraction by phase. An ensemble Kalman filter assimilating sparse, noisy wall measurements tracks the true trajectory and gauges how far the instantaneous wall shear determines the cycle: the model holds the climate, and the forecast error it incurs from the unobserved interior motions is bounded and concentrated in the active phase.

Keywords: wall turbulence, reduced-order modelling, sparse identification of nonlinear dynamics, radial basis functions, Mahalanobis metric, invariant measure

1 Introduction

The dynamics of near-wall turbulence are organised around a self-sustaining cycle (Waleffe 1997). Within this cycle, elongated low-speed streaks form by lift-up (Brandt 2014), undergo a sinuous instability, break down (Andersson et al. 2001), and regenerate over a characteristic time of about a hundred wall units (Hamilton et al. 1995). The cycle is spatially compact, confined to a minimal flow unit (Jiménez and Moin 1991), and it remains autonomous when the outer flow is removed (Jiménez and Pinelli 1999).

The discovery of this cycle motivated strategies to control it. These strategies are based on sensing the velocity close to the wall and acting on the wall to break the cycle and reduce drag, ranging from opposition control (Choi et al. 1994) to machine-learned policies (Guastoni et al. 2023; Cavallazzi et al. 2025). However, sensing at that wall-normal distance requires reaching into the flow above the wall, which is impractical in experiment and at high Reynolds number, whereas the shear stress at the wall itself is directly measurable (Naughton and Sheplak 2002; Löfdahl and Gad-el Hak 1999). The question becomes whether the cycle can be recovered from wall variables alone.

Recovering the signature of the near-wall structures from the wall has been approached in several settings, mainly wall modelling (Piomelli and Balaras 2002; Bose and Park 2018). Near-wall coherent motions (Robinson 1991) leave a footprint on the wall, in the pressure (Naguib et al. 2001) and in the shear stress (Kravchenko et al. 1993), so the near-wall field is observable from wall quantities alone and

can be reconstructed instantaneously (Güemes et al. 2021; Cuéllar et al. 2024). Larger structures further from the wall can be sensed too, provided they remain attached (Encinar and Jiménez 2019).

This footprint is uneven across the cycle, however: the wall-attached streaks leave a strong, direct signature in the streamwise shear (Robinson 1991), whereas the quasi-streamwise vortices of the breakdown and regeneration phase register only indirectly, through their imprint on the spanwise shear, which at the wall is the exact footprint of the near-wall streamwise vorticity, $\tau_y = \mu \omega_x$. An instantaneous reconstruction is a snapshot, though, not the cycle. Recovering the cycle’s dynamics, and so testing whether the wall retrieves it at all, needs more than a snapshot: a model that advances the wall state on its own, staying bounded and reproducing the correct statistics under free integration.

Modelling such dynamics from data is a task that can be handed to a reduced-order model, which compresses the flow into a few latent coordinates, far below the dimension of the underlying direct numerical simulation, and advances them in time. Such models are either intrusive or non-intrusive (Benner et al. 2015).

Intrusive models use knowledge of the governing equations, projecting them onto the latent space. POD–Galerkin models are the canonical example: proper orthogonal decomposition (Lumley 1967; Sirovich 1987) supplies the compression, and the Navier–Stokes equations are projected onto the retained modes, globally (Cavaleri and Nogueira 2022) or on locally defined bases (Colanera and Magri 2025, 2026). Since the truncated small scales drain energy from the retained modes, such projections are prone to instability (Rempfer 2000), and the discarded scales must be reintroduced through a calibrated eddy-viscosity dissipation to recover stable long-time statistics (Aubry et al. 1988).

Non-intrusive models instead learn the reduced dynamics directly from data, absorbing the truncation effect implicitly and requiring no access to the governing operator. Some remain opaque, such as recurrent networks that predict accurately but act as black boxes (Srinivasan et al. 2019); others are interpretable, such as sparse symbolic regression (Brunton et al. 2016) or cluster-based networks of coherent states (Kaiser et al. 2014; Fernex et al. 2021).

For the minimal channel, though, these models sit on one side of what is needed or the other. The reconstructions take the wall as input but carry no dynamics, returning only the instantaneous field (Guastoni et al. 2021); the reduced-order models that do carry the dynamics are built on the full field, not on the wall (Nakamura et al. 2021). A reduced-order model of the dynamics from the wall alone, which is what testing the cycle’s wall-recoverability asks for, does not yet exist.

Data assimilation bridges the two by folding measurements into a running model: an ensemble Kalman filter corrects the forecast towards partial, noisy observations of the full-order flow. Colburn et al. (2011) did so for a turbulent channel with the governing equations in the loop, and Özalp et al. (2026) with a data-driven reduced-order model in their place. The same filter has two uses: it pulls the model towards the true trajectory, and the mismatch between its forecast and the observations measures how much correction the model needs to stay consistent with the data (Desroziers et al. 2005; Anderson 2007).

We propose to test whether the wall closes the near-wall cycle as a dynamical object, using the reduced-order model of Pérez Cuadrado et al. (2026), whose machinery is taken as established and applied to a minimal turbulent channel (Jiménez and Moin 1991) at friction Reynolds number $Re_\tau = 180$ (Kim et al. 1987). The observable is the wall shear stress alone, which is the direct footprint of the near-wall velocity gradient and whose two components record the streaks and the quasi-streamwise vortices; the non-local wall pressure is left aside. The field is compressed by proper orthogonal decomposition, and its reduced coordinates are advanced by a vector field fitted to the data, without the governing equations.

This combination is new. Prior wall-based estimators are either intrusive, tied to the governing equations (Colburn et al. 2011), or rest on opaque data-driven models (Özalp et al. 2026), whereas the present model is non-intrusive, interpretable, and preserves the invariant measure of the wall shear (Vlachas et al. 2020). Measure preservation also aids the assimilation: a model that stays on the attractor gives the filter a physically consistent forecast covariance (Evensen 2003; Dee 1995), where a drifting model would report the wrong one.

Three results follow. Clustering the wall-shear trajectory recovers the coherent-structure skeleton: its phases fall onto the streak–vortex plane as a directed cycle, from the quiescent state through the vortex-dominated state to the mature streak, and the field fitted on this partition reproduces the invariant measure of the skin friction under free integration. Read as a dynamical system, that field places its slow points only in the quiescent phase and amplifies perturbations most as the streak matures towards breakdown. The ensemble Kalman filter then measures how complete the wall-only description is: the additive inflation needed to keep its forecast consistent with sparse, noisy wall sensors is the price, in inflation units, of the interior variability the wall cannot carry. The wall shear reproduces the cycle’s statistics on its own but not its trajectory, a statistical rather than deterministic autonomy of the cycle in the wall observable.

2 Modelling pipeline

The model advances the wall shear stress in a latent space, following the reduced-order pipeline introduced by Pérez Cuadrado et al. (2026), which we reproduce in Fig. 1 for reference. At each instant the two wall shear-stress components are encoded by proper orthogonal decomposition into a small set of latent coordinates. A vector field, fitted once to the data, advances those coordinates in time, and the decoder maps them back to the wall shear-stress fluctuation. The vector field is built from a radial-basis library whose centres respect the geometry of the attractor rather than the variance hierarchy of the decomposition.

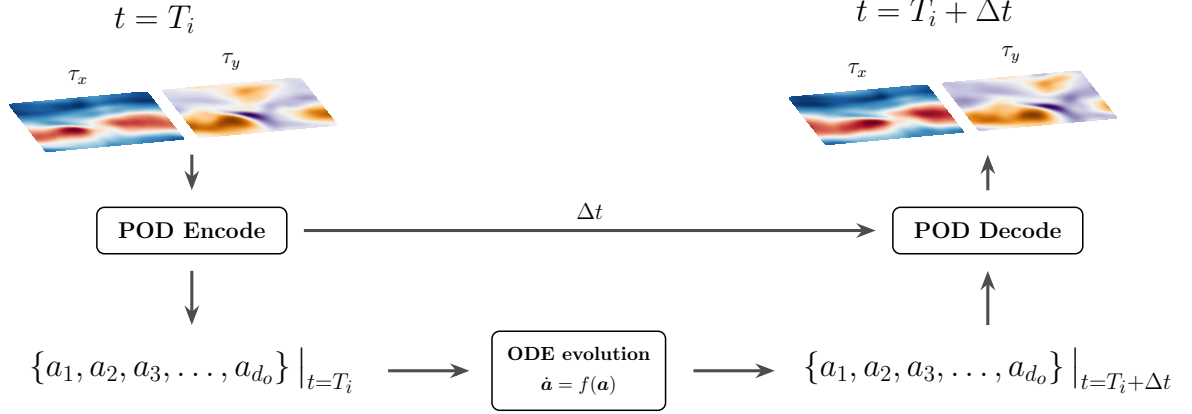


Fig. 1 Modelling pipeline. The wall shear-stress components (τ_x, τ_y) at time T_i are encoded onto latent coordinates by proper orthogonal decomposition, advanced by the learned sparse vector field $\dot{\mathbf{a}} = f(\mathbf{a})$, and decoded back to the wall shear-stress fluctuation at $T_i + \Delta t$.

The remainder of the paper follows this chain. Section 3 describes the channel simulation and the proper orthogonal decomposition of the wall shear. The latent trajectory is then clustered, and the resulting partition is shown to recover the coherent-structure skeleton of the near-wall cycle (Section 4). On the corrected geometry the sparse vector field is fitted and integrated in free running to test its invariant measure and predictability horizon (Section 5). The fitted field, explicit and differentiable, is then read as a dynamical system, giving the cycle’s slow points and the ordering of non-normal growth around the burst (Section 6). Sparse and noisy wall observations are finally folded back into the running model with an ensemble Kalman filter, whose required inflation bounds the interior variability the wall does not carry (Section 7).

3 Flow configuration and the wall observable

The wall shear stress that the model advances is produced by direct numerical simulation of a turbulent channel flow at friction Reynolds number $Re_\tau = 180$, computed with the solver CaNS (Costa 2018), which integrates the incompressible Navier–Stokes equations on a staggered grid by second-order finite differences in space, a low-storage three-stage Runge–Kutta scheme in time and a pressure-correction step that enforces continuity at each substep, with the flow held at constant mass flow rate.

The domain follows the minimal channel introduced by Jiménez and Moin (1991), a half-channel of streamwise, spanwise and wall-normal extent $480^+ \times 140^+ \times 180^+$, closed by a symmetry condition at the midplane. The wall-parallel plane is resolved by 64×64 points, giving spacings $\Delta x^+ \approx 7.5$ in the streamwise and $\Delta y^+ \approx 2.2$ in the spanwise direction, while the wall-normal direction is stretched, its near-wall spacing decreasing to $\Delta z^+ = 0.54$ at its minimum. This resolution is consistent with established near-wall channel simulation (Kim et al. 1987). Integration proceeds at a fixed step $\Delta t = 0.01$, and the two wall shear-stress components, the streamwise τ_x and the spanwise τ_y , are recorded every five steps, about half a wall time unit apart.

This interval is set by the accuracy of the reduced description rather than by storage: it must resolve the reduced coordinates finely enough for their time derivatives, which the model targets, to be estimated smoothly and for the projected coefficients to be well resolved in time. Sampling continues to 10^5 snapshots, spanning roughly 5×10^4 wall time units.

At each instant the two components (τ_x, τ_y) are stacked into a single vector $\mathbf{u} \in \mathbb{R}^N$ of dimension $N = 2 \times 64 \times 64 = 8192$. The $M = 10^5$ recorded fields are reduced by proper orthogonal decomposition (Lumley

1967; Sirovich 1987). The temporal mean is removed first, so that the basis describes the fluctuations about the mean wall state,

$$\bar{\mathbf{u}} = \frac{1}{M} \sum_{m=1}^M \mathbf{u}_m, \quad \mathbf{u}'_m = \mathbf{u}_m - \bar{\mathbf{u}}. \quad (1)$$

Collecting the centred fields as the columns of a snapshot matrix and taking its singular value decomposition, the equations

$$\mathbf{X} = [\mathbf{u}'_1, \dots, \mathbf{u}'_M] \in \mathbb{R}^{N \times M} \quad \text{and} \quad \mathbf{X} = \mathbf{\Phi} \mathbf{\Sigma} \mathbf{\Psi}^\top \quad (2)$$

give the proper orthogonal modes as the columns of $\mathbf{\Phi}$, an orthonormal basis ordered by the singular values on the diagonal of $\mathbf{\Sigma}$, whose squares measure the fluctuation energy carried by each mode. Retaining the leading d_o modes $\mathbf{\Phi}_{d_o}$, the reduced coordinates and the reconstruction of the wall field are

$$\mathbf{a}(t) = \mathbf{\Phi}_{d_o}^\top (\mathbf{u}(t) - \bar{\mathbf{u}}) \in \mathbb{R}^{d_o}, \quad \mathbf{u}(t) \approx \bar{\mathbf{u}} + \mathbf{\Phi}_{d_o} \mathbf{a}(t), \quad (3)$$

and the projection of each snapshot gives the reduced trajectory $\mathbf{a}_m = \mathbf{\Phi}_{d_o}^\top \mathbf{u}'_m$ on which the model operates.

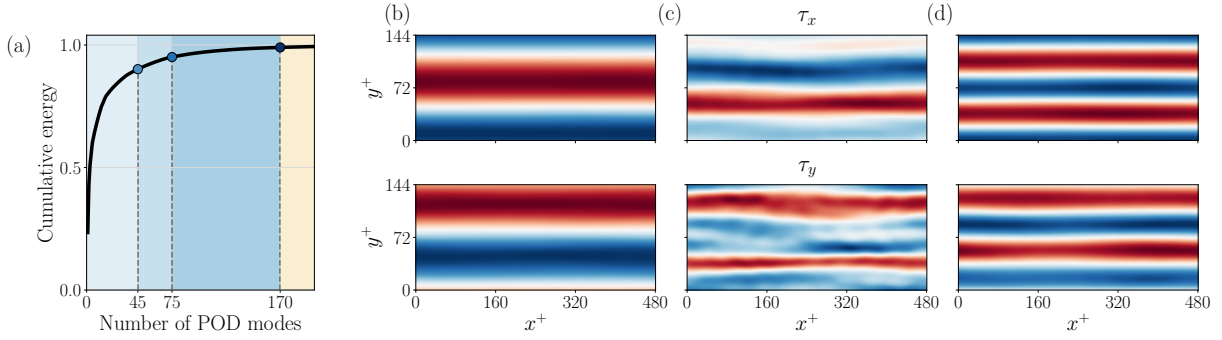


Fig. 2 Proper orthogonal decomposition of the wall shear stress. (a) Cumulative fluctuation energy against the number of retained modes, with the 90%, 95% and 99% thresholds reached at $d_o = 45$, 75 and 170 modes. (b–d) Streamwise τ_x (top row) and spanwise τ_y (bottom row) components of proper orthogonal modes 2, 4 and 6

The model operates on the leading $d_o = 45$ modes, which carry about 90% of the fluctuation energy (Fig. 2a). The discarded tail is dominated by small-scale, low-energy wall-gradient content, so the operating dimension follows the dynamically relevant scales rather than the reconstruction accuracy alone. That $d_o = 45$ carries the cycle’s dynamics is borne out by the free-run climate of Section 5, where the modes added up to $d_o = 75$ leave the recovered statistics unchanged; the clustering diagnostics of Section 4 independently share the same elbow at $d_o = 45$, 75 and 170. These higher-dimensional truncations are carried alongside for comparison.

The leading modes carry the wall imprint of the near-wall streaks; the streamwise and spanwise components of mode 2 are shown in Fig. 2(b). Modes 1 and 2 form a symmetric pair, and together they span the streak over its spanwise phase. This is consistent with velocity-field proper orthogonal decompositions of channel turbulence (Moin and Moser 1989; Aubry et al. 1988), whose wall imprint the shear-stress decomposition inherits.

The model requires the reduced coordinates together with their time derivatives. The derivatives are obtained by numerical differentiation of the coordinate time series, a five-point finite-difference stencil accurate to fourth order in the sampling interval (Brunton et al. 2016). Finite differencing amplifies high-frequency noise, but the retained coordinates are smooth, the small-scale tail having been truncated, so noise-resistant differentiation (Chartrand 2011) is not required. Each snapshot then carries the reduced state and its velocity as a pair $(\mathbf{a}_m, \dot{\mathbf{a}}_m)$, and this set of pairs is the data from which the reduced dynamics are fitted.

4 Dynamical states via unsupervised clustering

The latent trajectory is partitioned into clusters, each adapting the shape of its radial basis functions to the local geometry of the manifold (Pérez Cuadrado et al. 2026). The number of clusters is judged by

the separation it draws between neighbouring regions of the attractor, measured by three diagnostics: the marginal Bayesian information criterion (Schwarz 1978) $\Delta\text{BIC}/\Delta K$ (Fig. 3(a)), the shape difference between neighbours $|P_i - P_j|$ (Fig. 3(b)), and the relative speed gap $|v_i - v_j|/(v_i + v_j)$ (Fig. 3(c)).

Beyond $K = 16$ all three truncations level off, oscillating about the same range. The final values of the three metrics differ between $d_o = 45$, 75 and 170, but the curves share the same shape and the same elbow, a first indication that $d_o = 45$ already carries the skeleton of the dynamical system.

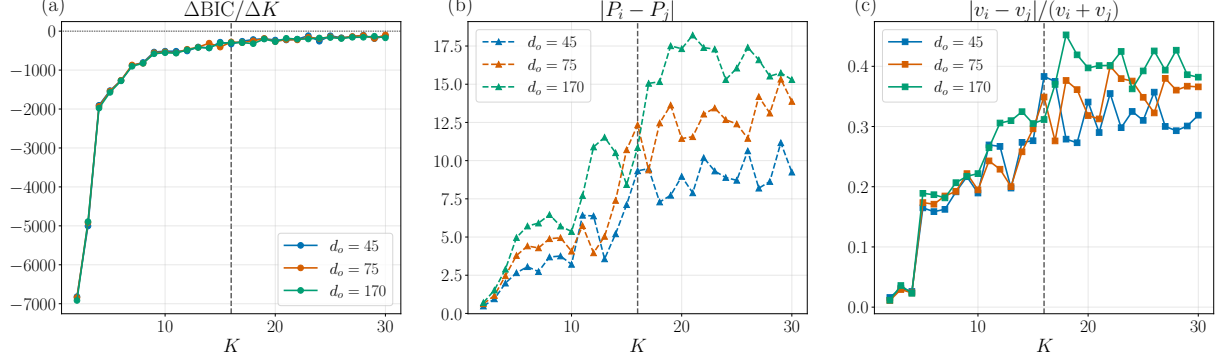


Fig. 3 Selecting the number of clusters. Three separation diagnostics against the number of clusters K , each for $d_o = 45$, 75 and 170, with the chosen $K = 16$ marked by the dashed line: (a) the marginal Bayesian information criterion $\Delta\text{BIC}/\Delta K$; (b) the participation-ratio gap $|P_i - P_j|$ between neighbouring clusters; (c) the relative speed gap $|v_i - v_j|/(v_i + v_j)$.

These diagnostics fix the number of clusters but not their meaning. Whether the sixteen clusters mark stages of the known wall cycle or merely cut the attractor into unconnected regions is settled by the cycle itself. If each phase is a distinct physical event, the wall signature of each cluster should carry information about the stage it represents. Since the expected alternation is between streaks and quasi-streamwise vortices, we look for the distinct scalar footprint each leaves on the wall.

The streaks are streamwise-elongated, spanwise-alternating modulations of the streamwise velocity (Robinson 1991), so τ_x is the natural streak marker. The streak intensity is the streamwise-averaged, $k_y = 1$ Fourier amplitude of the reconstructed fluctuation,

$$A_x = \frac{1}{\tau_w} \left| \widehat{\langle \tau'_x \rangle_x} (k_y=1) \right|, \quad (4)$$

with τ_w the mean wall shear, $\langle \cdot \rangle_x$ the streamwise average and the hat the spanwise Fourier transform. The streamwise average sets $k_x=0$, and the fundamental spanwise mode $k_y=1$ isolates the single-wavelength streak. This wall-shear amplitude is the analogue of the fundamental-spanwise-mode streak measure of Hamilton et al. (1995).

The vortices leave their mark on the spanwise shear instead: at the wall the spanwise shear is the exact footprint of the near-wall streamwise vorticity, $\tau_y = \mu \omega_x$, so its strength measures the vortices directly. The vortex strength is taken as the root-mean-square of that fluctuation over the wall plane,

$$\sigma_y = \frac{1}{\tau_w} \langle \tau_y'^2 \rangle^{1/2}. \quad (5)$$

The two markers place each wall field on the streak–vortex plane (A_x, σ_y) , where the cycle’s alternation between streak formation and vortex regeneration should appear as an orbit and each cluster as one of its phases.

The cluster-averaged markers (standardised to zero mean and unit variance) are shown in Fig. 4(a), together with the dominant transitions between clusters. The occupied quadrants mark three phases of the regeneration cycle (Hamilton et al. 1995; Waleffe 1997): the quiescent state, low A_x and low σ_y , in Q3; the mature high-amplitude streak in Q1; and the vortex-dominated state, σ_y high with the coherent streak still weak, in Q2. This interpretation is coherent with the cluster-averaged τ_x fields in Fig. 4(d–f): Q1 carries a strong, streamwise-coherent low-speed streak; Q2 the same bands with their streamwise coherence reduced, modulated along the streamwise direction and pinched into closed τ_x contours; and Q3 a weak, low-amplitude remnant of the streak.

The partition used no physical labels, yet its centroids fall onto the three phases of the near-wall cycle once read through the wall markers, each phase in a cluster of its own. The separation follows from the

manifold-adapted clustering of Pérez Cuadrado et al. (2026), which corrects the coordinate and sampling biases that would otherwise blur the phases together.

The transitions between these phases carry a directed sense, made visible in Fig. 4(b) by high-pass filtering the two markers and rendering their mean phase velocity as streamlines. The spanwise shear leads the streak by about $13t^+$, roughly a tenth of the cycle, so the circulation is clockwise, the vortices rising ahead of the coherent streak before both markers relax towards the quiescent Q3, the order consistent with lift-up (Brandt 2014).

The circulation is not a causality claim. Built from wall footprints, it is the kinematic skeleton of the wall-projected dynamics, the phases and the sense in which they succeed one another rather than the drive behind them. On a closed cycle a phase lead marks position, not drive, so the observed lift-up sense need not follow the causal arrow; the dominant causal direction found by Lozano-Durán et al. (2020) in fact runs the other way, the streak instability regenerating the vortices. The wall thus returns the cycle’s directed skeleton, not the mechanism that drives it.

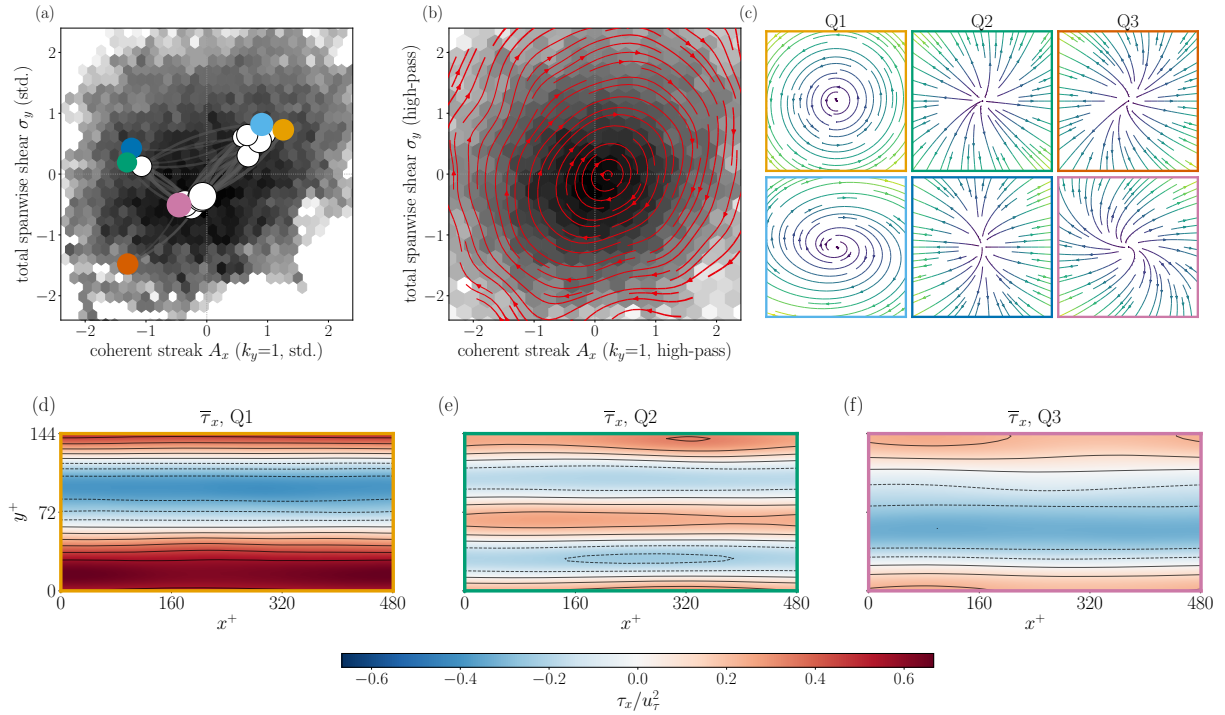


Fig. 4 The near-wall cycle on the streak–vortex plane. (a,b) Plane spanned by the coherent streak amplitude A_x , the $k_y = 1$ spanwise fundamental of τ'_x , and the spanwise-shear intensity σ_y . (a) The sixteen cluster centroids on the standardised plane, sized by occupancy, over the sampled density (grey), with arrows for the dominant inter-cluster transitions. (b) The same plane after high-pass filtering, with red streamlines of the mean phase velocity; the winding directedness $D = \bar{\theta}/|\bar{\theta}| = -0.20$ gives a weak but significant clockwise circulation. (c) Local dynamics of the fitted field, the leading-plane projection of the Jacobian J_k at two clusters per quadrant, streamlines shaded by the local rate; Border colours link each tile to its cluster in (a). (d–f) Cluster-averaged streamwise shear $\bar{\tau}_x$ of the representative cluster in Q1, Q2 and Q3: the mature streak, its loss of coherence, and the quiescent state.

The clusters fall on the cycle’s phases, but position is not dynamics: a partition can carve a continuous attractor into pieces that sit apart yet behave alike. Clustering is justified only where the local dynamics, not the position alone, separate the phases (Pérez Cuadrado et al. 2026). Being differentiable, the fitted field lets this be checked at each cluster’s centroid: linearising it there gives the Jacobian J_k , which reads out how the trajectories passing through the cluster are deformed.

At each centroid the leading eigenvalue of J_k has positive real part, so the leading direction expands and pushes nearby trajectories apart. The character of that eigenpair sets the phases apart (Fig. 4c): the mature streaks of Q1 are complex-dominant, a rotational deformation of neighbouring trajectories; the Q2 clusters largely real-dominant, straining them without rotation; Q3 lies between. This split confirms that the clusters isolate dynamically distinct stages of the cycle. Where cluster-based reduced-order models represent the phase dynamics by transition probabilities between clusters (Kaiser et al. 2014; Fernex et al. 2021), the differentiable field here lets each phase be read instead as a local deformation.

5 Model fitting and free integration

After clustering and adapting the RBFs to the local geometry (Pérez Cuadrado et al. 2026), the model is fitted by a single sparse regression over the radial basis functions, giving the vector field $\dot{\mathbf{a}} = f(\mathbf{a})$. The record is split in time, the first 80% used for the regression and the final 20% held out to test it.

Two parameters of the library are left free: the number of basis functions N and the tangent anisotropy α_{tan} , which scales the width of every kernel along the local manifold tangent by a single common factor. The fit is studied at two truncations, $d_o = 45$ and 75 . The modes beyond these are small-scale wall-gradient fluctuations, incoherent with the streaks and quasi-streamwise vortices that carry the cycle (Jiménez and Moin 1991; Hamilton et al. 1995; Waleffe 1997), so the 99%-energy truncation $d_o = 170$ of Section 4 is not carried into the dynamical model. Both parameters are swept for $d_o = 45$ and 75 (Fig. 5a,b); the error shown is the test NRMSE, normalised per mode by that mode’s test-segment standard deviation and averaged over the retained modes.

At small α_{tan} the kernels are narrower than the spacing between their centres and their supports leave gaps across the manifold; coverage then improves only slowly with N , and at the smallest widths no number of kernels within the swept range fills the space, so the error stays near unity. As α_{tan} grows the supports overlap, the manifold is covered, and the error decreases. Once the width is sufficient each curve reaches a minimum at an intermediate N and rises beyond it, the fit passing from under-parameterised to over-parameterised as the extra kernels begin to follow noise; the larger the width, the fewer kernels needed to reach that point.

The floor of each curve, its lowest error over N , is shown against α_{tan} in Fig. 5(c), where the two truncations cross. At small α_{tan} the lower-dimensional model reaches the lower floor: with fewer modes the spread of characteristic scales across dimensions is narrower, so a single global α_{tan} fits every dimension well. Past $\alpha_{\text{tan}} \approx 10$, where coverage is no longer the constraint, the higher-dimensional model takes the lead: retaining more modes leaves less energy in the truncated tail, and that smaller cut-off loss sets the error.

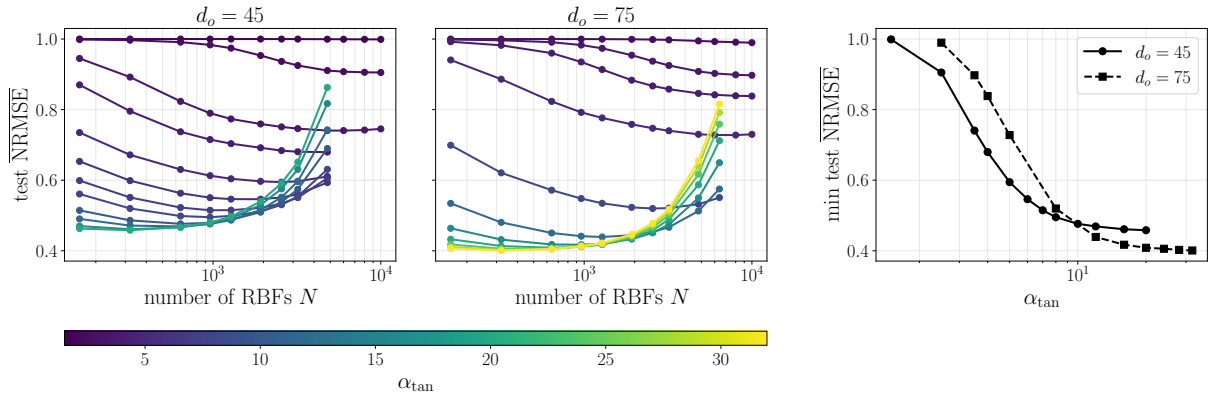


Fig. 5 Fitting the sparse vector field. Mode-averaged test NRMSE against the number of radial basis functions N , one curve per tangent anisotropy α_{tan} (colour), for (a) $d_o = 45$ and (b) $d_o = 75$. (c) The NRMSE floor over N against α_{tan} .

With the library fixed, the fitted field is advanced by a fourth-order Runge–Kutta scheme at the sampling step on which it was fitted, $\Delta t \approx 1.13 t^+$, together with the per-cluster trust-region corrector of the method paper (Pérez Cuadrado et al. 2026), calibrated as described there. Each cluster’s trust region is set to enclose the sampled attractor, so on-attractor trajectories never trigger the corrector and the reported statistics are those of the fitted field itself; the corrector acts only on excursions beyond the sampled data, pulling the trajectory back to keep the run bounded.

The free run is judged on two indicators. The first measures short-range accuracy through the valid prediction time (VPT), a standard gauge of forecast quality (Vlachas et al. 2020): how long a forecast stays close to the reference. It is the time in t^+ before the reconstruction error e_{rec} , the latent-space error normalised by $\sigma_A = (\sum_i \text{Var}(a_i))^{1/2}$, the per-mode standard deviations over the reference record summed in quadrature, first exceeds 0.25; that is,

$$\text{VPT} = \min\{t : e_{\text{rec}}(t) > 0.25\}, \quad e_{\text{rec}}(t) = \frac{\|\mathbf{a}_{\text{ROM}}(t) - \mathbf{a}_{\text{true}}(t)\|}{\sigma_A}. \quad (6)$$

The normalisation is global, by the single σ_A rather than per mode, so the threshold weights the energetic modes that carry the wall signature; the horizon is reported in wall units t^+ .

The long-time match is read from two distances between the free run and the reference, both standard metrics for comparing distributions (Gibbs and Su 2002). With $s(t) = \langle \tau_x \rangle(t)$ the instantaneous skin friction and F its empirical distribution over the free-run time series, the first is the Wasserstein-1 distance (Peyré and Cuturi 2019),

$$W_1 = \int |F_{\text{ROM}}(s) - F_{\text{FOM}}(s)| ds, \quad (7)$$

reported normalised by the reference standard deviation σ_{FOM} . The second compares the shape of the skin-friction spectrum. Writing $\hat{\phi}(f)$ for the premultiplied power spectral density $f P(f)$ of each series, normalised to unit area over the same resolved band $f \leq f_{\text{max}}$, above which the reference spectrum is band-limited, the two shapes are compared by the total-variation distance between them as probability densities over frequency (Gibbs and Su 2002),

$$\text{TV} = \frac{1}{2} \int |\hat{\phi}_{\text{ROM}}(f) - \hat{\phi}_{\text{FOM}}(f)| df. \quad (8)$$

The half-factor and unit-area normalisation bound $\text{TV} \in [0, 1]$, zero for identical spectral shapes and one for disjoint support; being area-normalised it measures shape alone, complementing W_1 . Both spectra are estimated with the same Welch parameters on a common frequency grid, so the distance is free of binning artefacts.

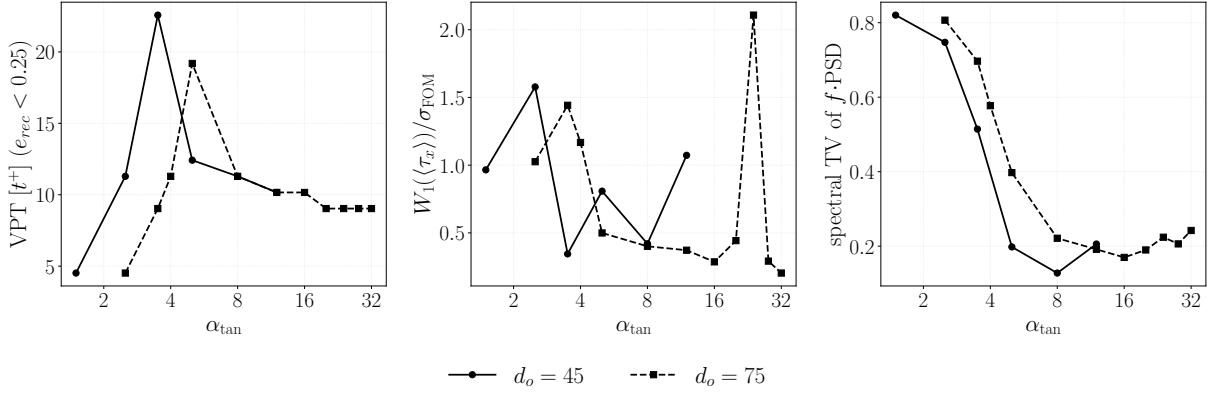


Fig. 6 Predictability and climate against anisotropy. For $d_o = 45$ and 75 against α_{tan} : (a) valid prediction time to reconstruction error $e_{\text{rec}} = 0.25$; (b) Wasserstein-1 distance of the skin-friction distribution in units of σ_{FOM} ; (c) total-variation distance of the premultiplied spectrum $f \cdot \text{PSD}$.

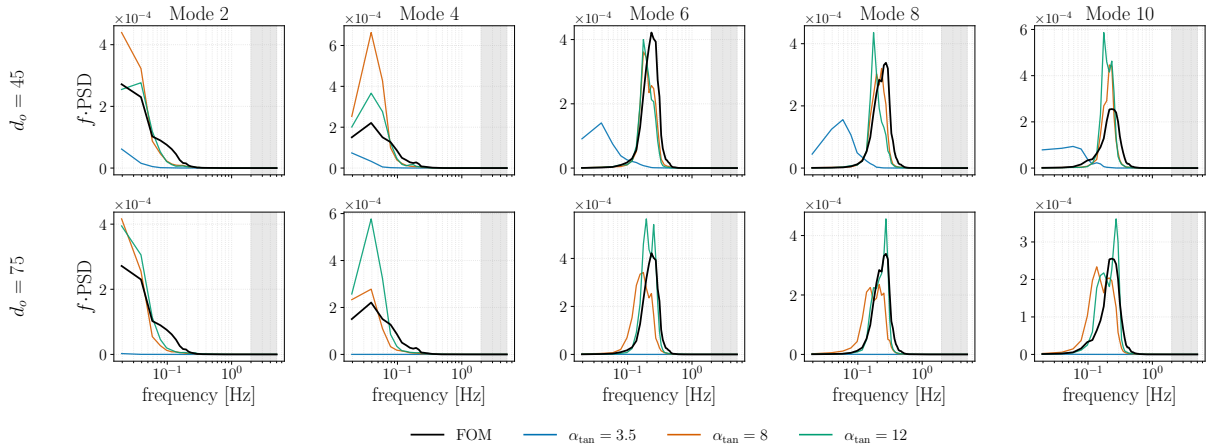


Fig. 7 Spectral fidelity of the free run. Premultiplied spectra $f \cdot \text{PSD}$ of latent modes 2, 4, 6, 8 and 10 for the free-integrated model at three values of α_{tan} and the reference (FOM), for $d_o = 45$ (top) and 75 (bottom). Shading marks frequencies above the reference band limit.

The three indicators are read against α_{tan} for both d_o in Fig. 6. As expected from the gap between one-step fit and closed-loop rollout, the valid prediction time, averaged over 100 initial conditions spaced uniformly across the test segment, about one regeneration cycle apart (panel a), does not follow the NRMSE floor of Fig. 5(c): the fit keeps improving with α_{tan} , whereas the VPT peaks at an intermediate width, $\alpha_{\text{tan}} \approx 3.5$ to 5, and declines beyond it, in both d_o . The peak width is unchanged when the error threshold is varied from 0.2 to 0.35, so the ranking does not hinge on the tolerance.

The climate does not follow the VPT either. A larger W_1 means a wider gap from the reference skin-friction distribution, and its minimum lies at a larger α_{tan} than the VPT peak (panel b); the spectral distance behaves the same way (panel c). The one-step fit, the predictability and the long-time statistics are each optimised at a different width, so a single global α_{tan} trades one against another.

The origin of this trade-off is the plane average itself. It keeps only the spatial-mean part of each mode, so $\langle \tau_x \rangle$ is carried by a few low-order modes; the energetic streak pair, spanwise-alternating, averages to near-zero across the span and barely contributes. At small α_{tan} the coverage gaps leave the fitted field unable to drive even these modes, and the free run relaxes towards a near-fixed state, so $\langle \tau_x \rangle$ is weak (Fig. 8). As the width grows the modes regain their motion, but around $\alpha_{\text{tan}} \approx 3.5$ to 5 the active low-order modes oscillate more slowly than the reference (Fig. 7), so the reconstructed $\langle \tau_x \rangle$ is red-shifted rather than lacking its faster content.

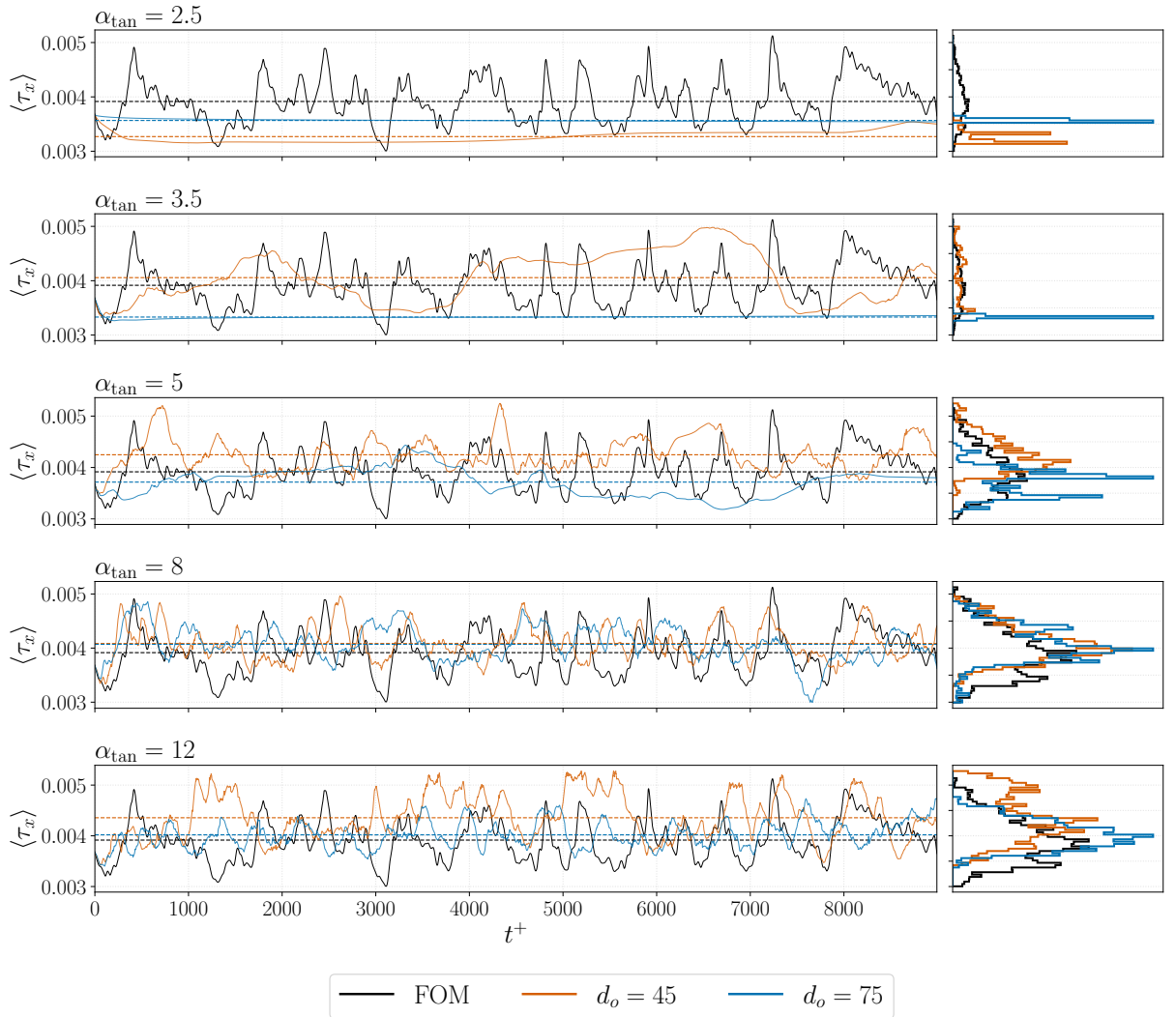


Fig. 8 Free-running skin friction. Spatially averaged streamwise wall shear $\langle \tau_x \rangle$ against time for the free-integrated model at $d_o = 45$ and 75 and the reference (FOM), one row per α_{tan} , with the time means dashed. Right: the marginal distributions.

The climate, by contrast, sharpens at larger widths, the friction and spectral distances reaching their lowest values around $\alpha_{\text{tan}} \approx 8$ to 16. Wider, more strongly overlapping kernels smooth the fitted field: this costs local accuracy, and so predictability, but favours a bounded, correctly occupied long-time state,

reminiscent of the trade of short-term accuracy for long-time stability seen when dissipation is added to POD–Galerkin models (Aubry et al. 1988; Rempfer 2000), here a by-product of the wider kernels rather than an explicit dissipation.

The higher truncation does not shift this: on all three measures $d_o = 75$ is no better than $d_o = 45$. The modes added between the two are already inert for the skin-friction statistics targeted here, and this is the evidence for not fitting the still smaller-scale tail up to $d_o = 170$: more of that incoherent content adds reconstruction accuracy without enriching the dynamics.

These trade-offs set the operating point. A wall-only model is asked to carry the cycle’s statistics rather than a single trajectory, so the width is chosen for the invariant measure: the model is run at $d_o = 45$ and $\alpha_{\text{tan}} = 8$, the low end of the range where the friction distribution and the spectrum are closest to the reference, kept near the predictability peak, at the cost of a shorter prediction horizon than the width that maximises it. At this setting the free run reproduces the skin-friction distribution and premultiplied spectrum of the direct simulation (Fig. 8, Fig. 7). The wall-shear field, integrated on its own, sustains the near-wall cycle’s statistics.

6 Reading the near-wall cycle from the fitted field

The same field that sustains these statistics is explicit and differentiable, so it can be read as a dynamical system rather than only run forward: its resting states, its Jacobian, and the growth of small perturbations are all available in closed form. This is what the present section exercises; it closes on the limit of what the wall state can determine.

Since the field $\dot{\mathbf{a}} = f(\mathbf{a})$ is defined over the whole attractor, its slow points can be located directly. A slow point is a state at which the model’s tendency vanishes, an equilibrium of the fitted field near which the flow moves slowly. The local dynamics there are unstable (Section 4), so these are weakly unstable near-equilibria.

Solving $f(\mathbf{a}) = 0$ for the production model, every recovered slow point lies in the quiescent half of the cycle ($\sigma_y < 0$), spread across streak amplitude, reminiscent of the lower- and upper-branch family of invariant solutions of wall-bounded shear flow (Nagata 1990; Waleffe 2001; Toh and Itano 2003) (Fig. 9). The roots sit well within the sampled data, where the corrector never acts, so they are not artefacts of thin kernel coverage.

The same holds across four independently fitted models, whose escape-time distributions agree (Fig. 9b). The cycle’s rhythm is asymmetric: the quiescent phase offers weakly unstable anchors the flow lingers near for tens of t^+ , while the active phase holds nothing and is crossed quickly. The near-wall cycle’s unstable invariant solutions show the same rhythm, shadowed between fast bursts (Kawahara and Kida 2001; Kawahara et al. 2012); these slow points of the wall-visible mean are not those solutions, nor equilibria of the flow, which has none.

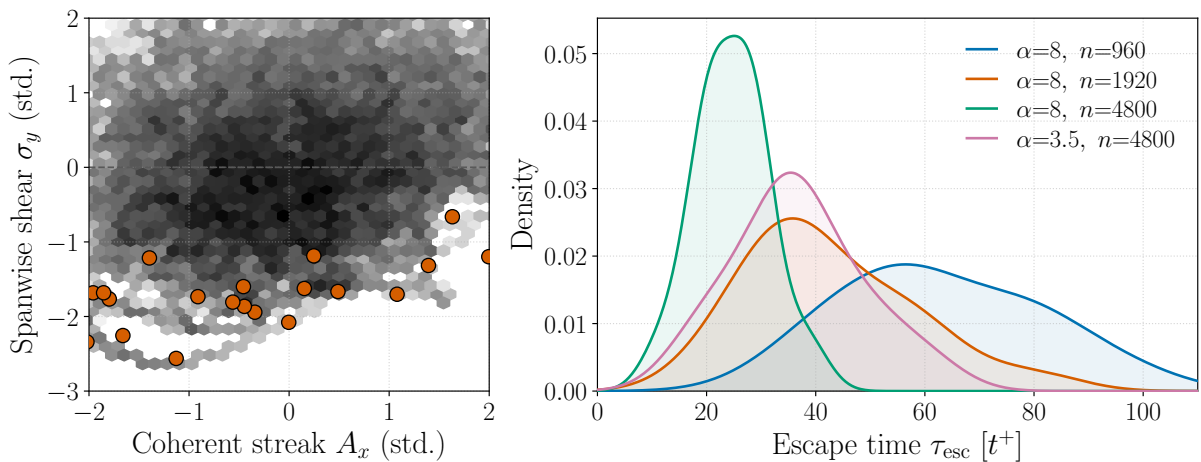


Fig. 9 Slow points live only in the quiescent phase. (a) States at which the fitted field’s mean tendency vanishes, for the production model ($\alpha_{\text{tan}} = 8$, 1920 kernels), on the streak–vortex plane over the sampled density (grey); all lie in the quiescent half ($\sigma_y < 0$), the active half none. (b) Escape-time distribution of the slow points for four independently fitted models, varying α_{tan} and kernel count n : tens of t^+ in every case.

The slow points say where the flow rests; the Jacobian says how nearby trajectories are stretched and squeezed. Differentiating the field gives $J = \partial f / \partial \mathbf{a}$, and linearising it reads off the local anatomy of the cycle (Fig. 10). Beyond the eigenvalues that separated the phases in Section 4, its non-normality measures the worst-case transient amplification of a disturbance over a short horizon, and its trace whether a cloud of nearby states contracts.

The non-normal growth asks how sensitive the flow is to perturbations. Over a $10t^+$ horizon, G is the largest factor by which a small disturbance grows in the latent-coefficient energy norm, the leading singular value of the tangent-linear propagator integrated along the trajectory rather than frozen at a point (Schmid 2007; Trefethen et al. 1993). Built this way, it captures the transient amplification a non-normal field allows even where the eigenvalues decay (Butler and Farrell 1992). A large G marks a sensitive phase, where a small difference grows fast; read per cluster (Fig. 10a), it peaks at burst onset, where a nudge grows about 2.5-fold. This is the growth in the wall-visible coordinates alone, and the latent norm is not the physical energy, so G is a wall-projected proxy for the full growth rather than a bound on it.

The trace of J looks at a whole cloud of nearby states rather than a single disturbance, and indicates the rate at which a small volume of states contracts or grows. As shown in Fig. 10a, the squeezing is hardest deep in the burst and slackens to almost nothing at burst onset. The sign of $\text{tr } J$ there varies between fits, so only the ordering is read, and it is the contraction of the wall-visible latent volume, not the flow’s own dissipation.

Placed on the cycle of Section 4, the most amplifying phase is the mature streak (Q1): the streak is at its most fragile just as it reaches full strength, on the point of breaking down. The quiescent state (Q3) is the least sensitive of all. The strongest contraction sits in the breakdown itself, spanning the late streak and the vortex-dominated phase, where the disturbances the streak has amplified are dissipated as it comes apart.

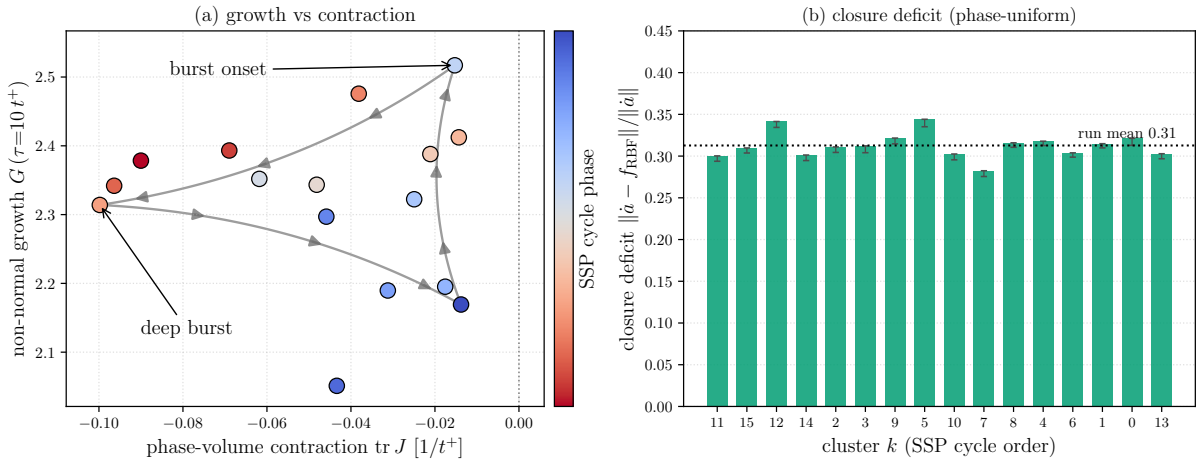


Fig. 10 Dynamical anatomy of the cycle, read off the fitted field. (a) Non-normal growth G , the worst-case amplification over a $10t^+$ horizon, against phase-volume contraction $\text{tr } J$, one point per cluster, coloured by cycle phase. Burst onset is least contracting and most amplifying (upper right), the deep burst most contracting; G is the wall-visible amplification, reported against the field’s modal and numerical-abscissa bounds, and $\text{tr } J$ is the mid of the four-fit range. (b) Closure deficit per cluster: the fraction of the wall tendency not determined by the wall state, near 0.3 and phase-uniform.

The readings so far are what the wall field reveals; one number bounds what it cannot determine (Fig. 10b). About 30% of the instantaneous wall tendency cannot be written as a function of the wall state, the same in every phase. This is the share of what happens next that the wall cannot know from what it sees now, set by the interior motions it does not observe, and it is why the model holds the climate but not the trajectory. The next section measures what this one-step deficit accumulates to over the assimilation window, in inflation units.

7 Data assimilation and wall-observable autonomy

The free run of Section 5 reproduces the cycle’s statistics, and with them its invariant measure. The near-wall region sustains its cycle when the motions above the buffer layer are quenched (Jiménez and Pinelli 1999), which justifies a low-dimensional model of it; whether the wall shear, a two-component footprint

Table 1 Baseline data-assimilation parameters.

Parameter	Symbol	Value
Ensemble size	N_{ens}	100
Sensor stride	Δ	8
Observed components		τ_x, τ_y
Number of sensors	N_{obs}	128
Observation noise	σ_{obs}	0.1
Assimilation cadence	Δt_{obs}^+	5
Multiplicative inflation		1.02
Additive inflation	q	0.1

of that three-dimensional cycle, is a sufficient coordinate for it is a separate question. That the climate is correct does not mean the instantaneous wall shear fixes what the cycle does next.

The cycle turns on the buffer-layer state above the wall: the wall-normal velocity that drives the lift-up, and the streamwise vortices that regenerate the streaks (Waleffe 1997; Hamilton et al. 1995). The wall carries only their footprint, so a field fitted in wall coordinates learns a conditional-mean tendency (Chorin et al. 2000), averaged over the interior motions it does not see. Section 6 read this off as a closure deficit at a single step; what follows measures the forecast error it produces over short windows, the cost the wall-only model pays for the events off the wall.

Data assimilation is the natural tool: the wall shear is what an experiment can measure, even with sparse and noisy sensors, and the forecast is corrected online, without re-solving the model or forming an adjoint, so it is cheap. Assimilating wall measurements into a flow model has precedent with the governing equations in the loop (Colburn et al. 2011) and with a data-driven reduced model in their place (Özalp et al. 2026). The estimator is a Kalman filter, alternating a forecast, in which the model advances the state and its uncertainty, and an analysis, in which the forecast is pulled towards the observations, the two weighted by their covariances through the Kalman gain (Kalman 1960).

The classical filter propagates the full covariance through the linearised dynamics. To avoid the linearisation, an ensemble Kalman filter (EnKF) carries a Monte Carlo ensemble, integrates each member through the nonlinear field, and reads the forecast covariance from their spread (Evensen 2003). This captures the nonlinear error growth along the attractor, at a cost that scales with the ensemble size and the latent dimension rather than the full state.

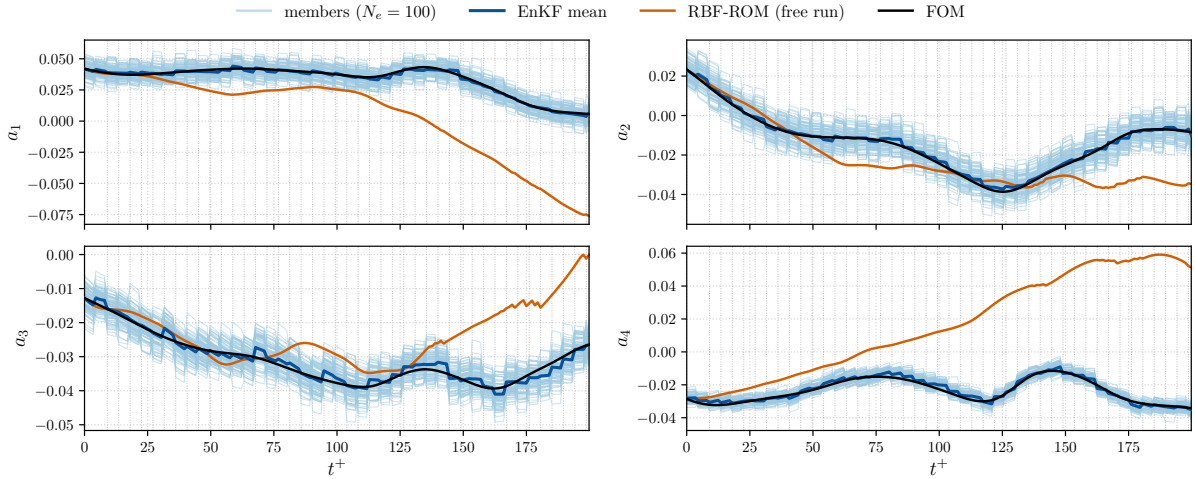


Fig. 11 Ensemble forecast of the leading modes. Latent coefficients a_1 to a_4 against t^+ for the ensemble Kalman filter ($N_e = 100$ members), its mean, the free-running RBF-ROM and the reference (FOM). Dotted verticals mark the analysis times.

The filter runs on the reduced coordinates, observing the two wall-shear components on a uniform grid of stride Δ , so $N_{\text{obs}} = 2(64/\Delta)^2$ per sample. Each sensor carries independent Gaussian noise of standard deviation σ_{obs} relative to the root-mean-square of the wall-shear fluctuation, giving a diagonal observation covariance, and the observations are perturbed per member. An ensemble of N_{ens} members is initialised by perturbing a snapshot about the attractor, advanced through the fitted field between

analyses, and corrected every Δt_{obs}^+ . Multiplicative and additive inflation counter ensemble collapse, the additive level q being the knob examined below; Table 1 lists the baseline values, with the stride, noise level and cadence varied later. On the leading modes (Fig. 11) the ensemble brackets the reference, and its mean tracks the true trajectory where the free run, given the same start, drifts away.

At each analysis the filter compares the incoming observations with its own forecast; the mismatch between the two, the innovation d , is the only quantity it can weigh against its own error estimate. Scaling d by the covariance $S = \mathbf{H}\mathbf{P}^f\mathbf{H}^\top + \mathbf{R}$ it predicts for it gives the normalised innovation squared, a measure of whether the mismatch exceeds the spread the filter expected. Here $\mathbf{H}$ is the linear map from the reduced state to the sensors, $\mathbf{P}^f$ the forecast covariance carried by the ensemble spread, and $\mathbf{R}$ the diagonal observation-noise covariance.

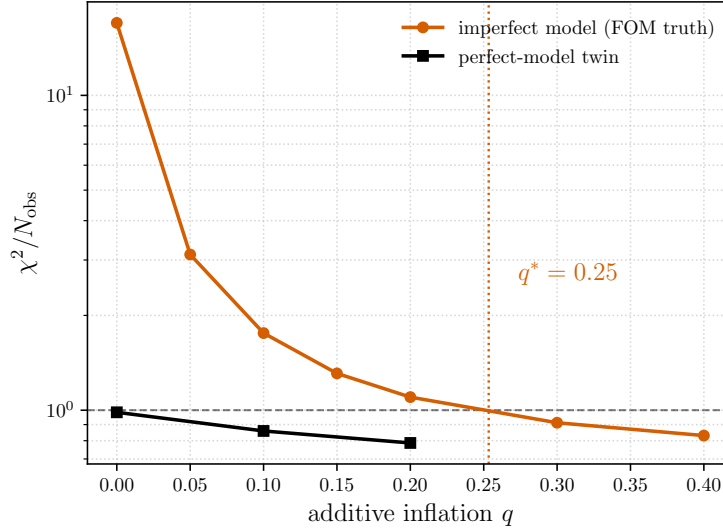


Fig. 12 The additive-inflation floor. Time-averaged χ^2/N_{obs} against the additive inflation q , for the model assimilating FOM observations and for the perfect-model twin. The crossing of $\chi^2/N_{\text{obs}} = 1$ (dashed) sets q^* ; the twin needs none.

When the filter is consistent this statistic averages to one per observation, $\chi^2/N_{\text{obs}} = 1$. This is known as the standard innovation-consistency check (Desroziers et al. 2005).

If the model that advances the ensemble is missing dynamics, it produces innovations too large for the spread it predicts, so $\chi^2/N_{\text{obs}} > 1$. Enlarging the forecast covariance by additive inflation q restores consistency, and the level that brings χ^2/N_{obs} back to one is q^* : the process noise the wall-only model must inject to account for its own error, the established way of reading model error off the innovations (Dee 1995; Anderson 2007; Tandeo et al. 2020).

The value of q^* carries a physical reading: it is an upper bound on the forecast error the wall-only model incurs from the interior motions it does not carry, in units of the attractor’s climatological standard deviation. Figure 12 locates it as the crossing of $\chi^2/N_{\text{obs}} = 1$. A perfect-model twin, whose truth is generated by the model itself, needs none: its q^* is zero, so the floor measured against the true flow is model error and not an artefact of the filter. For the baseline configuration $q^* = 0.25$: over each $5t^+$ interval the model must inject process noise of amplitude one-quarter of the climatological standard deviation, a few per cent of the climatological variance, to stay statistically consistent with the observations.

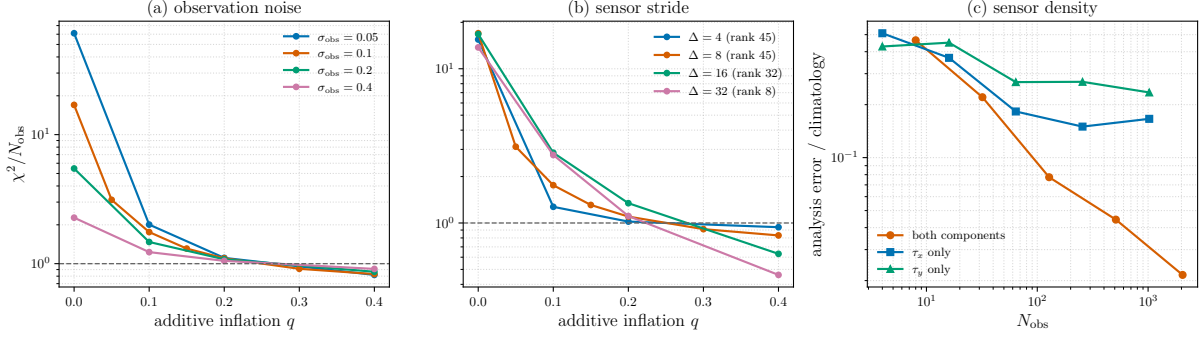


Fig. 13 q^* is a property of the model. χ^2/N_{obs} against q across (a) observation-noise level σ_{obs} and (b) sensor stride Δ (rank of $\mathbf{H}$ in the legend); the unit crossing barely moves. (c) Analysis error, normalised by climatology, against sensor count N_{obs} for τ_x , τ_y and both components.

To separate the model-error floor from the way the wall is observed, q^* is measured across the measurement configuration: the observation-noise level and the sensor stride (Fig. 13). Over an eightfold range of noise (a) and of stride (b), the crossing of $\chi^2/N_{\text{obs}} = 1$ barely moves, holding near $q^* \approx 0.25$. Raising the noise only lowers the curve at $q = 0$, since a larger $\mathbf{R}$ makes the same innovations less surprising, and leaves the crossing where it was. The stride is more severe: at $\Delta = 32$ the wall resolves only 8 of the 45 modes, the rank of $\mathbf{H}$ in the legend, yet q^* is unchanged. The floor is therefore a property of the model, not of the filter or the sensing, which is what lets it be read as the model’s own error.

The state estimate itself does depend on the sensing. Panel (c) shows the analysis error, normalised by the climatological spread, falling as sensors are added; both shear components together beat either alone, and τ_x carries more than τ_y , the streaks leaving a stronger streamwise footprint than the quasi-streamwise vortices leave in the spanwise shear. Denser, two-component sensing thus sharpens the estimate without moving the floor: how well the wall is read and how much the wall omits are separate quantities.

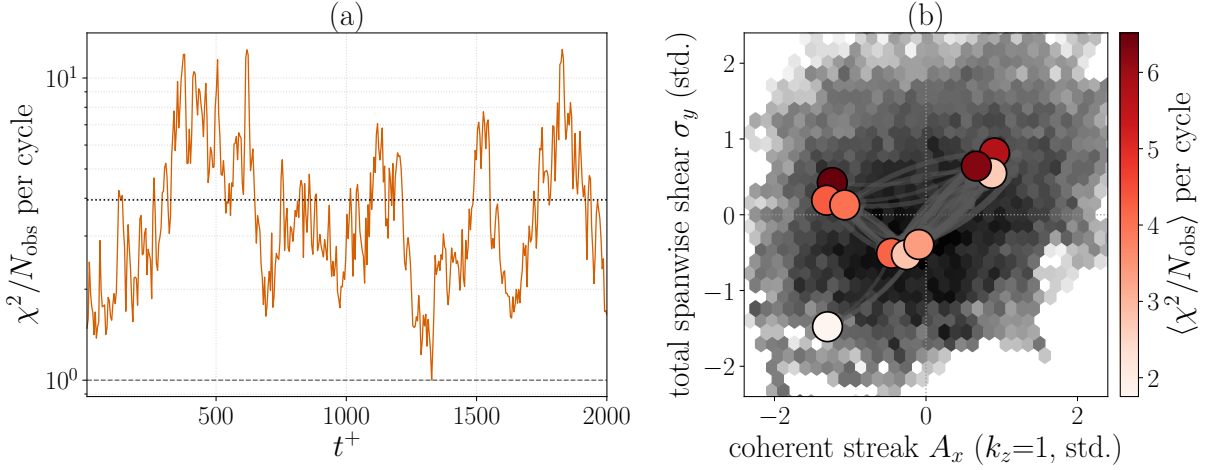


Fig. 14 Innovation consistency in time and around the cycle. (a) Per-cycle χ^2/N_{obs} over a 2000 t^+ run; the cycle mean (dotted) sits above consistency (dashed), the excess arriving in bursts. (b) The same statistic cycle-averaged per cluster, on the streak–vortex plane over the sampled density (grey) and coloured by $\langle \chi^2/N_{\text{obs}} \rangle$; it is larger in the active, high- σ_y phase and near-consistent in the quiescent phase, tracing the cycle (faint arrows).

The floor q^* is a time average, and resolved along the run it is not uniform. As shown in Fig. 14a, the per-cycle χ^2/N_{obs} is bursty and the excess arrives in bursts. When mapped per cluster along the cycle and streak and spanwise wall shear metrics, the innovation is largest in the active half (Fig. 14b), spanning the mature streak and its breakdown. However, on the quiescent phase the average reduced significantly.

This excess is genuine forecast error rather than ensemble under-dispersion: the realised error is larger in the active phase while the spread stays flat across the cycle, and it scales with the cycle phase, not the state amplitude.

The one-step closure deficit of Section 6 is phase-uniform, so this localisation is a matter of its integration: the fitted field amplifies a perturbation most in the same active phase, where the interior

motions the wall cannot resolve, the lift-up and the streak breakdown that regenerates the vortices, are also strongest. The model’s missing information is thus concentrated in the phase where the cycle is driven from off the wall.

8 Conclusions

The near-wall self-sustaining cycle has been reduced to a differentiable latent field fitted from wall shear stress alone. Resampling the trajectory in arc length and clustering under a local Mahalanobis metric align the basis with the attractor’s geometry rather than the variance hierarchy of the decomposition, and under this geometry the mature-streak, breakdown and quiescent phases separate into sharp populations without physical labels. Integrated freely, the model stays bounded and reproduces the cycle’s statistics and its invariant measure, the climate and not any single orbit upon it.

Since the fitted field is explicit and differentiable, it can be read as a dynamical system. Its slow points are located by Newton iteration and lie in the quiescent phase, weakly unstable near-equilibria reminiscent of the cycle’s lower- and upper-branch invariant solutions. Its Jacobian resolves the anatomy of the cycle by phase: the non-normal growth of a disturbance peaks as the streak reaches full strength and breaks down, and the phase-volume contraction is strongest in the breakdown itself. These readings come directly from the fitted coefficients, without recourse to the governing equations.

Assimilating wall measurements into the model quantifies how far the wall determines the cycle. The wall-only model reproduces the climate but not the instantaneous trajectory: an additive-inflation floor of about a quarter of the climatological standard deviation, stable across the sensor noise and spacing, measures the forecast error it incurs from the interior motions it does not carry. This deficit is a fixed fraction of the tendency at each step and concentrates, once integrated, in the active phase where the lift-up and the streak breakdown are governed from off the wall. A field fitted in wall coordinates therefore learns a conditional-mean tendency, and these results are for a single friction Reynolds number in a minimal channel.

Several directions follow. The interior-information deficit invites a closure model that injects the unresolved off-wall dynamics to lower the floor and extend the horizon over which the wall-only forecast stays reliable. Raising the friction Reynolds number and enlarging the domain would test whether the wall stays a near-sufficient coordinate once the near-wall cycle is no longer isolated, and adding wall pressure to the shear would close more of the deficit at source. With sparse, noisy wall sensors the estimator already tracks the true trajectory, so the same construction feeds the near-wall control strategies that the problem began with.

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